XRPL Validator — Daily Engineering Report · 2026-04-02

Period: 2026-04-02 00:00 to 24:00 US Eastern (2026-04-02 04:00 to 2026-04-03 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: git history only. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 11 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **perf: parallel object fetching — 32 concurrent RPC requests per ledger** (`2241766`, xrpl-validator) Replace sequential ledger_entry fetch loop with tokio semaphore-bounded parallel requests. 500 objects: 1.5s sequential → ~0.2s parallel.
- **perf: bump concurrent RPC fetches from 32 to 64** (`4dd589f`, xrpl-validator) Round time: 1.2-1.4s → 0.8-1.2s. Fetch overhead now ~0.1s. Bottleneck is pure hash computation (~1s), not network.
- **perf: depth-4 caching (65536 buckets) — hash compute 1s → 15ms** (`ef3a806`, xrpl-validator) Replace 256 depth-2 buckets with 65536 depth-4 buckets. Only ~500 of 65536 buckets are dirty per ledger (0.8%) vs 230 of 256 (90%) before.
- **feat: Prometheus /metrics endpoint + stale cache fix** (`02c7f15`, xrpl-validator)
- Rename /metrics HTML page to /dashboard - New /metrics returns Prometheus text exposition format (14 metrics) - Metrics: uptime, ledger_seq, consecutive_matches, total_matches/mismatches, round_time, compute_time, peers, state_objects, ready_to_sign, txs, info labels - Delete disk cache on invalidate_tree() to prevent stale cache scan loops - ValidatorMetricsSnapshot struct for clean data assembly
- **feat: SQLite history storage + /api/history endpoint** (`931e666`, xrpl-validator)
- New history.rs module with HistoryStore backed by rusqlite (bundled SQLite) - Records every verified ledger round: seq, matched, healed, txs, objs, round_time_ms, compute_time_ms, peers, timestamp - /api/history endpoint returns 1h/24h/7d/30d summaries + recent entries - Wired into ws_sync: records after each push_sync_log call - Delete disk cache on invalidate_tree() to prevent stale cache scan loops - Growth: …
- **feat: public validator page + incident log** (`77a7fda`, xrpl-validator)
- /validator: polished public-facing landing page with live metrics, uptime chart, hash verification status, architecture overview - /incidents: operational transparency page with historical incident timeline - Caddy config

ready for validator.halcyon-names.io subdomain - Both pages pull from /api/state-hash and /api/history for live data - Clean corporate design with Inter font, responsive layout

- **feat: dashboard redesign, consistent nav, compute_time fix** (`8a71356` , xrpl-validator)
- Restore original SSE-driven metrics page with smooth sparklines - Transaction Types moved to full-width bottom row with stable layout - All known tx types shown permanently (inactive dimmed at 30%) - Consistent nav across all pages: Home | Consensus | Peers | State | Metrics | GitHub - Root / serves public validator page - Old feed moved to /feed - Fix compute_time_secs not being stored in update_and_hash status - …
- **fix: revert TLS cert/key mismatch from rejection to debug log** (`bddbb3f` , xrpl-validator) XRPL peers intentionally use ephemeral TLS keys separate from their node identity Public-Key. This is by design — authentication happens via session signatures, not TLS certificate pinning. The rejection was blocking all peer connections.
- **fix: metrics accuracy — engine hit rate shows -- when no txs applied, TLS check debug-only** (`f59d65b` , xrpl-validator)
- Engine hit rate: show "--" instead of "100%" when total_applied=0 - XRP Burned label: "All Time" → "Lifetime" for clarity - TLS cert/key mismatch: debug log only (XRPL uses separate TLS and protocol keys by design)

Docs and operations

- **security: fix critical and high audit findings** (`69835ca` , xrpl-validator)
- **security: fix all remaining audit findings (15 items)** (`bedb027` , xrpl-validator)

Independence program

No independence record for UTC day 2026-04-02; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
04-02 21:18	—	xrpl-validator	`2241766`	perf: parallel object fetching — 32 concurrent RPC requests per ledger
04-02 21:43	—	xrpl-validator	`4dd589f`	perf: bump concurrent RPC fetches from 32 to 64
04-02 22:57	—	xrpl-validator	`ef3a806`	perf: depth-4 caching (65536 buckets) — hash compute 1s → 15ms
04-02 23:50	—	xrpl-validator	`02c7f15`	feat: Prometheus /metrics endpoint + stale cache fix
04-03 00:17	—	xrpl-validator	`931e666`	feat: SQLite history storage + /api/history endpoint
04-03 00:22	—	xrpl-validator	`77a7fda`	feat: public validator page + incident log
04-03 02:07	—	xrpl-validator	`8a71356`	feat: dashboard redesign, consistent nav, compute_time fix
04-03 02:23	—	xrpl-validator	`69835ca`	security: fix critical and high audit findings
04-03 02:33	—	xrpl-validator	`bedb027`	security: fix all remaining audit findings (15 items)
04-03 02:41	—	xrpl-validator	`bddbb3f`	fix: revert TLS cert/key mismatch from rejection to debug log
04-03 03:33	—	xrpl-validator	`f59d65b`	fix: metrics accuracy — engine hit rate shows -- when no txs applied, TLS check debug-only

Times in UTC. A commit appears on the day it was committed to main or merged into it.